

Workshop

How to make a Research Map

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EAAD, UMinho, 2026-07-08



University of Minho
School of Architecture, Art and Design



Laboratory of Landscapes,
Heritage and Territory



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Construction
& Technology



CASAIIS

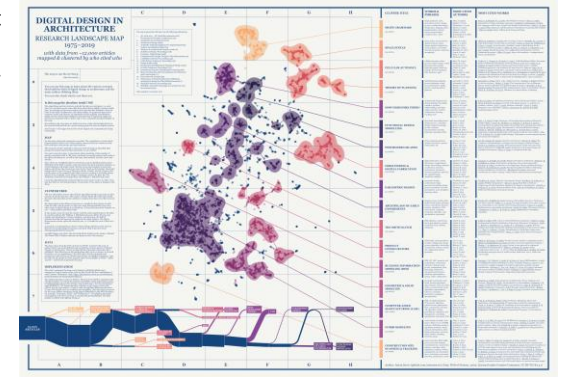
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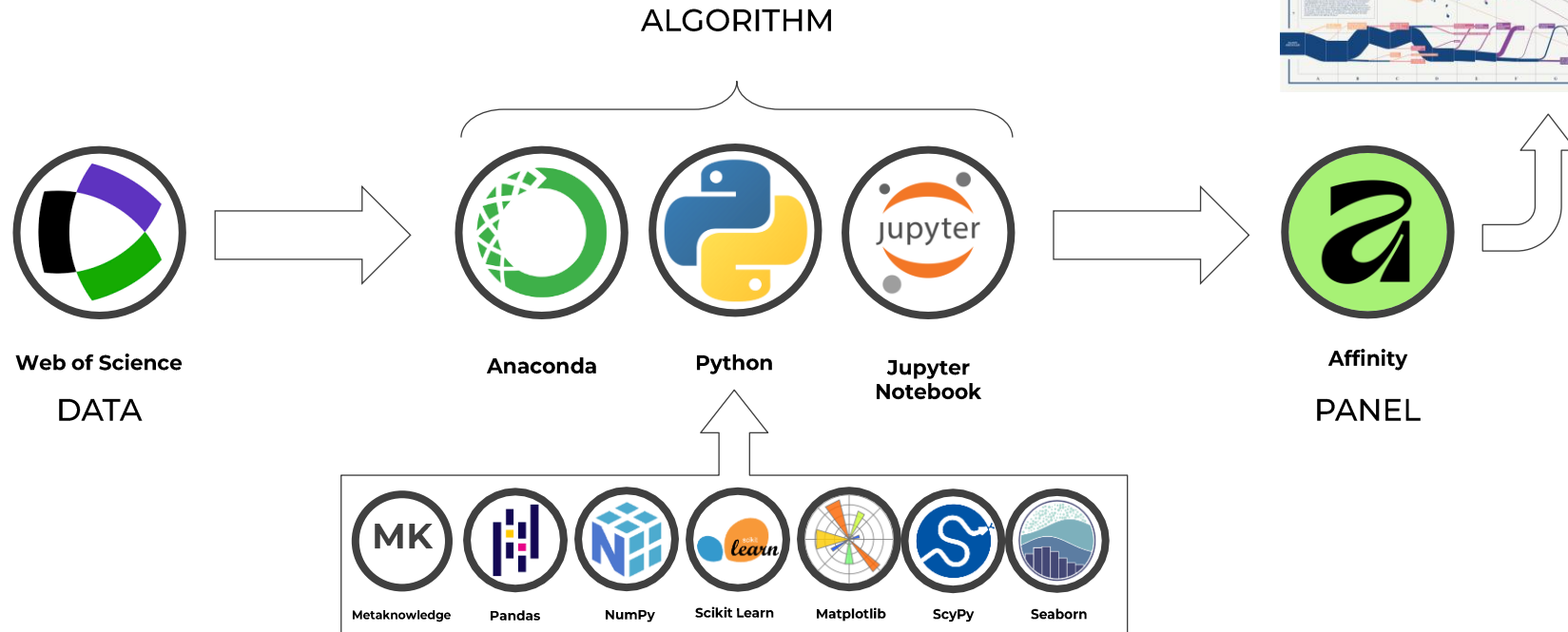
INTRODUCTION

Methodology

Reference:
Anton Savov (2020)
@ TU Darmstadt



REFERENCE



Web of Science (WoS)



What is WoS?

It is a paid platform, owned by Clarivate, that hosts indexed scientific articles and acts as a database of knowledge.

It can be used to search and read articles, as well as export their data for analysis.

At UMinho we have access to their collections while using the university's network, either directly or through VPN.



What is Python?

Python is a high-level, general-purpose programming language that emphasizes code readability, simplicity, and ease-of-writing with the use of significant indentation,[38] an extensive ("batteries-included") standard library, and garbage collection. Python supports multiple programming paradigms but with an emphasis on object-oriented programming and dynamic typing.

We can quickly and easily add functionality to Python by installing distribution packages from the internet. One such way to do download and manage them, as well as the version of Python itself, is with Anaconda.

source: [https://en.wikipedia.org/wiki/Python_\(programming_language\)](https://en.wikipedia.org/wiki/Python_(programming_language))



What is conda?

Conda is an open-source package and environment manager that installs, updates, and removes packages with their dependencies in isolated environments, allowing multiple versions of the same software without conflicts.

Anaconda is a software tool that bundles Python and conda, and is also the name of the company that manages it. Miniconda is a lightweight version of Anaconda.

Both Anaconda Distribution and Miniconda include conda, but they differ in what's pre-installed.

Feature	Anaconda	Miniconda
Created by Anaconda	Yes	Yes
Has conda	Yes	Yes
Has Anaconda Navigator	Yes	No
Packages included	600+	300+
Install space required	~9.7 GB	~900 MB

source: <https://www.anaconda.com/docs/getting-started/concepts/anaconda-or-miniconda>



What is Jupyter?

Jupyter Notebook is a notebook authoring application, under the Project Jupyter umbrella. Built on the power of the computational notebook format, Jupyter Notebook offers fast, interactive new ways to prototype and explain your code, explore and visualize your data, and share your ideas with others.

A computational notebook is a shareable document that combines computer code, plain language descriptions, data, rich visualizations like 3D models, charts, graphs and figures, and interactive controls. A notebook, along with an editor like Jupyter Notebook, provides a fast interactive environment for prototyping and explaining code, exploring and visualizing data, and sharing ideas with others.

source: <https://jupyter-notebook.readthedocs.io/en/latest/>

INSTALLATION

Step-by-Step Installation

1. Download Project Files → “.yaml” and “.ipynb” files
2. Install Anaconda → From anaconda.com (miniconda version recommended)
3. Create conda environment → With “conda create --name my-env --file rm_env.yaml” from CMD
4. Activate environment → With “conda activate my-env” from CMD
5. Launch Jupyter notebook → With “jupyter notebook” from CMD
6. Open Code → Open the .ipynb file in the jupyter notebook tab that opened in your browser

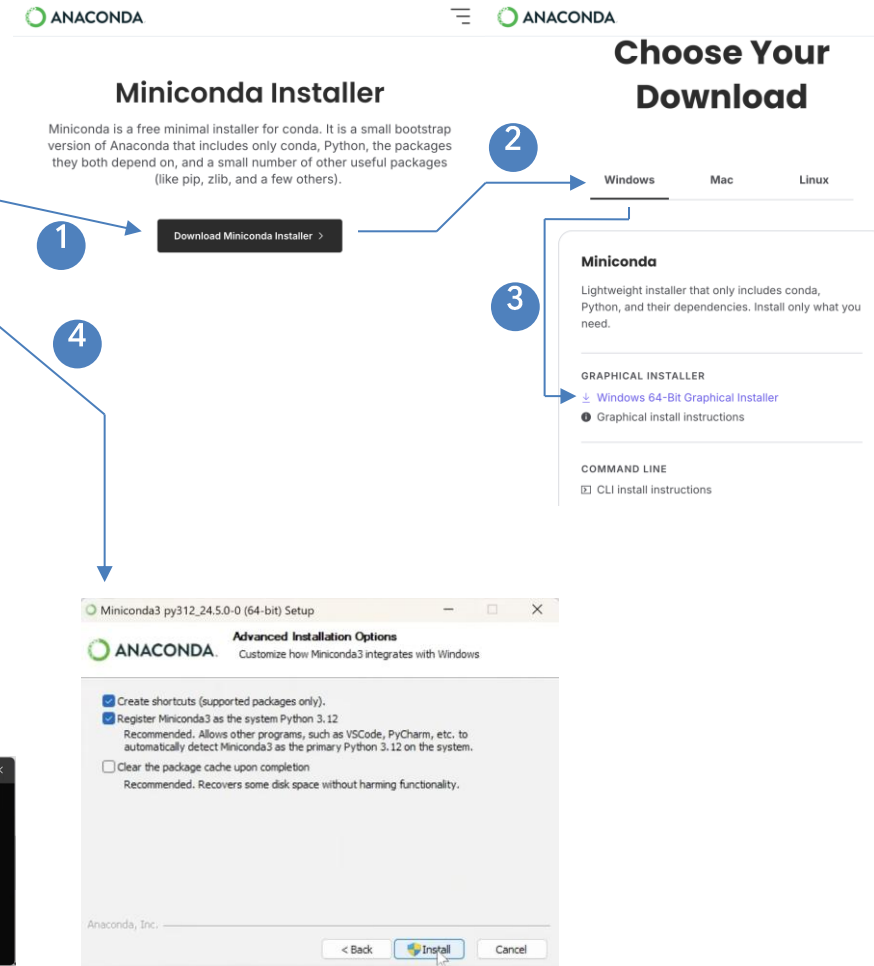
* Replace my-env with the name of the environment you want to create/activate.

1. Download Project Files

- I. Create a project folder in an easy to find place on your computer to store all the data
- II. Go to https://github.com/rodrigochiesse/research_map
- III. Download these files from the GitHub repository to your project folder:
 - rm_env.yaml
 - ResearchMap_RC.ipynb
- IV. continue

2. Installing Anaconda

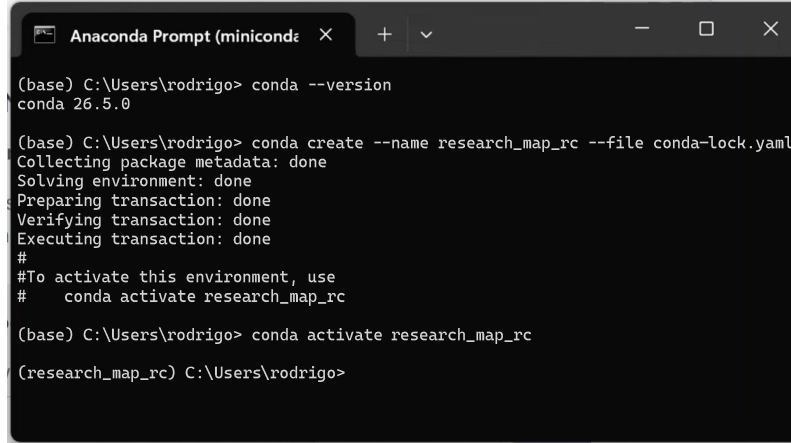
- I. Go to <https://www.anaconda.com/download>
- II. Scroll down to find “Download Miniconda Installer” and choose the installer for your OS
- III. Open the file and follow install instructions
- IV. Open Anaconda Prompt as Administrator
- V. continue



3 - 4. Creating and Activating Environment

- I. To test the install, in Anaconda Prompt, type:
conda --version
if the output is “conda” followed by a number, the install was successful
 - II. To create the environment from file, type:
conda create --name research_map_rc --file rm_env.yaml

 - III. Now activate the environment by typing:
conda activate research_map_rc
- Change this for your file path. Example:
C:\Users\rodrigo\folder\rm_env.yaml



```
Anaconda Prompt (miniconda) x + - □ x
(base) C:\Users\rodrigo> conda --version
conda 26.5.0

(base) C:\Users\rodrigo> conda create --name research_map_rc --file conda-lock.yaml
Collecting package metadata: done
Solving environment: done
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
#To activate this environment, use
#  conda activate research_map_rc

(base) C:\Users\rodrigo> conda activate research_map_rc

(research_map_rc) C:\Users\rodrigo>
```

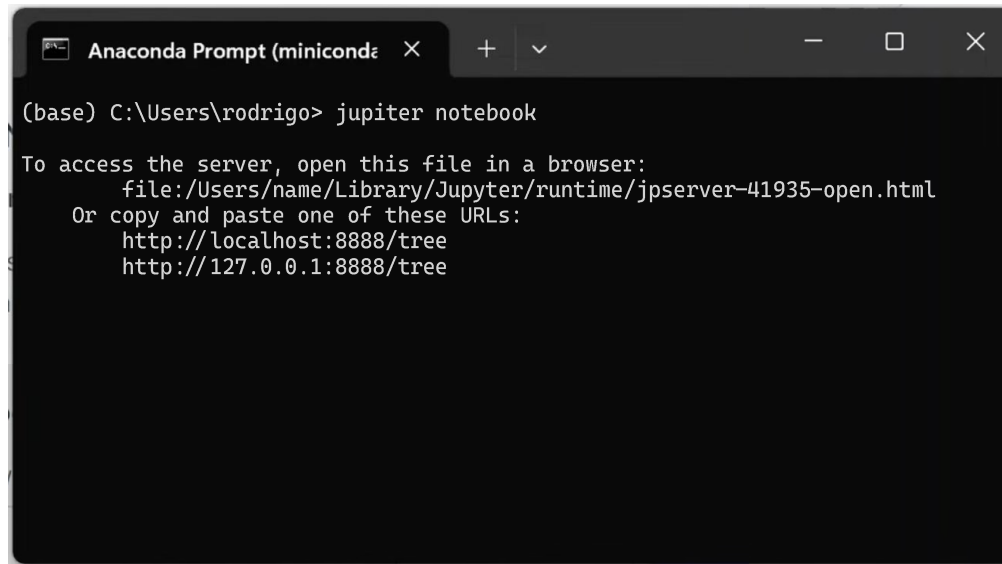
5 - 6. Launch Jupyter Notebook & Open Code

I. Type `jupyter notebook`

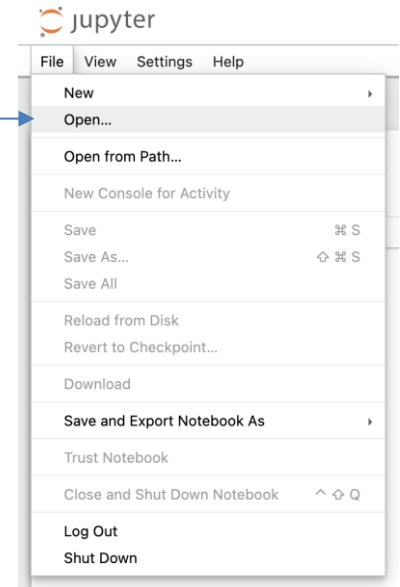
if your folder is in another directory, append `--notebook-dir D:/`

II. If a web browser window opened, great, if not, paste the address from the Terminal in your web browser

III. In the Jupyter notebook interface on your browser, go to **File>Open** and choose the file **ResearchMap_RC.ipynb** from your project folder



```
Anaconda Prompt (minicond...  
(base) C:\Users\rodrigo> jupyter notebook  
  
To access the server, open this file in a browser:  
file:/Users/name/Library/Jupyter/runtime/jpserver-41935-open.html  
Or copy and paste one of these URLs:  
http://localhost:8888/tree  
http://127.0.0.1:8888/tree
```

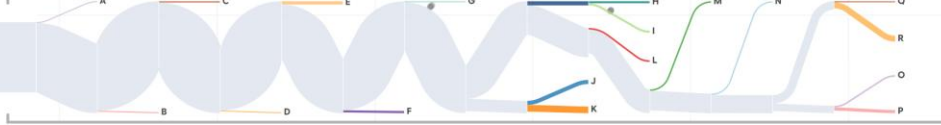


DATA GATHERING

EXPLAINING THE MAP

WITH DATA FROM ~15K ARTICLES MAPPED AND
CLUSTERED BY WHO CITED WHO

Licensed under Creative Commons: CC BY-NC-SA 4.0
Based on the work by Anton Savov (2020)
Data from Web of Science
By Rodrigo Chiesse

[illegible]

DIGITAL DESIGN IN ARCHITECTURE

RESEARCH LANDSCAPE MAP 1975-2026

WITH DATA FROM ~15K ARTICLES MAPPED AND
CLUSTERED BY WHO CITED WHO

Licensed under Creative Commons: CC BY-NC-SA 4.0

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INFORMATION

The algorithms used to structure and cluster the data are stochastic/non-deterministic, this means that every time the program runs the result is slightly different. The selection of literature might not be the absolute definitive research body in this field. Furthermore, the cluster labels are generated from the most common words and phrases, and most cited authors and works, so important but less popular contributions might not have been acknowledged. Literature that is important might not have been associated with any cluster and also remained unacknowledged on the map.

Nevertheless, the map provides a macro overview of the relationships between the various fields and can be a good starting point for more in-depth research.

MAP

On the map, each point represents an article. There are 15379 points, 7660 of them were classified as noise/outliers. The coordinates of each article depend on how close it is to other articles, measured by how similar their citations are. The points are sized based on the number of times the article has been cited. They also have transparency, so more intense the color, the denser the cluster, and stronger the relationship between the articles.

The research clusters marked in color are derived using an algorithm that groups the points based on their proximity to one another. For each research cluster, a label was generated consisting of a title, the most commonly occurring phrases and words in their articles' titles and abstracts, as well as the most cited authors and articles.

The most commonly occurring phrases are, in addition to single words, trigrams and bigrams.

The list of top-cited authors reveals who defines and drives the research in each cluster. Such as Branko Kolarevic being central to the 'Foundations of Digital Design in Architecture' cluster.

Finally, the list of most cited articles gives us the seminal works that everyone working in that field always refers to. For example, in the 'Shape Grammars' cluster, there are quite a few works by George Stiny.

I parsed the algorithmically extracted info for each cluster through an LLM (Gemini Pro 3.1) to give each of them a title.

CLUSTER TREE

The tree chart (Sankey diagram) below traces when the algorithm found necessary to split a group of articles. The sooner a cluster broke off of the main trunk, the more disconnected the articles it contains are from the rest of the research body.

Differently from Savov, I didn't observe the 'Shape Grammars' cluster being the first to split from the bulk, but I do agree, from a 'human analysis' standpoint, that it is the most distinguishable topic from every other topic in this map.

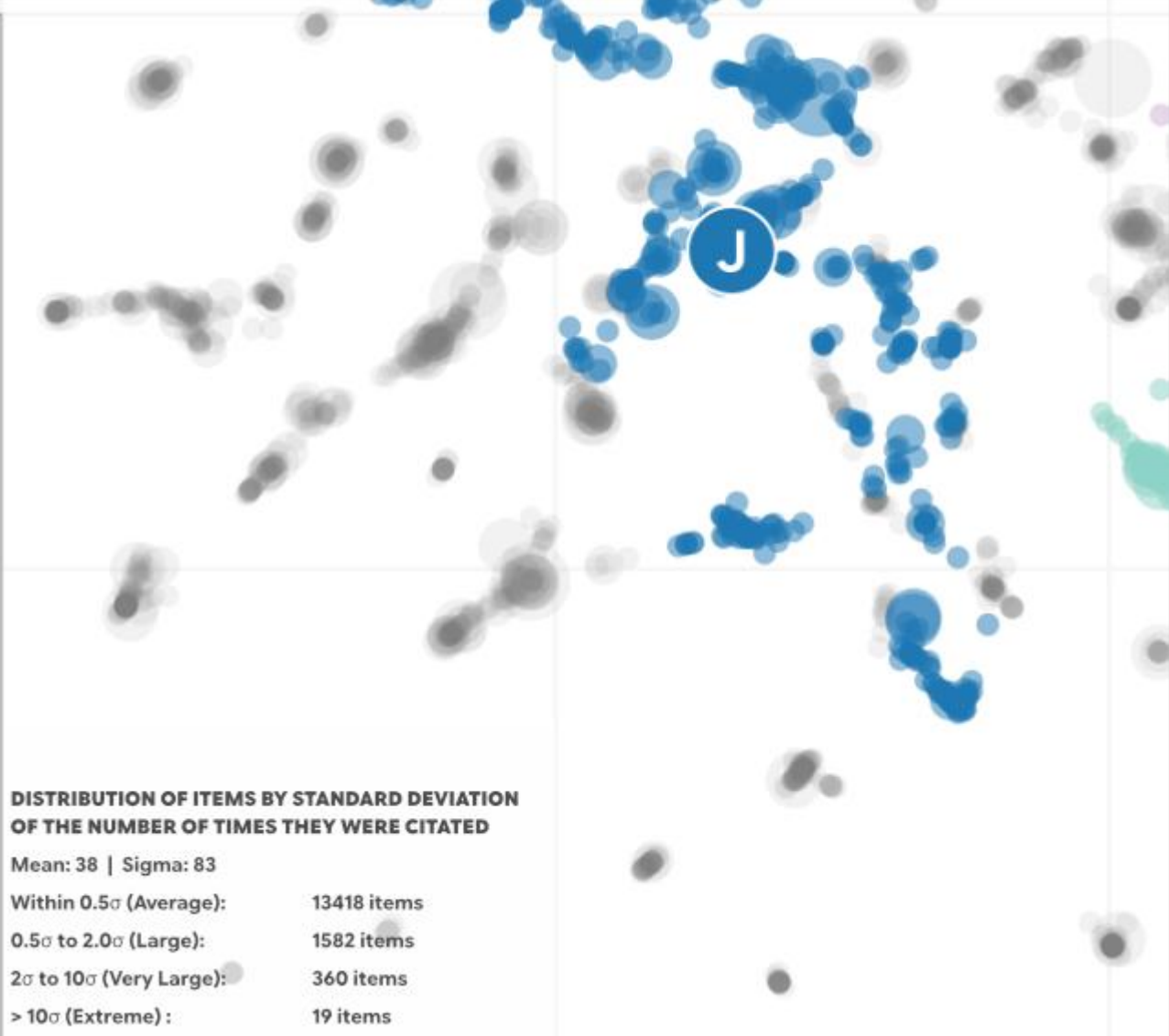
It also makes sense to track the order the clusters split along with the amount of citations, this way we can see what builds upon what, and can observe how the 'Shape Grammars', 'Design Thinking' and now 'AI' and 'Computer Vision' carry the bulk of the citations.

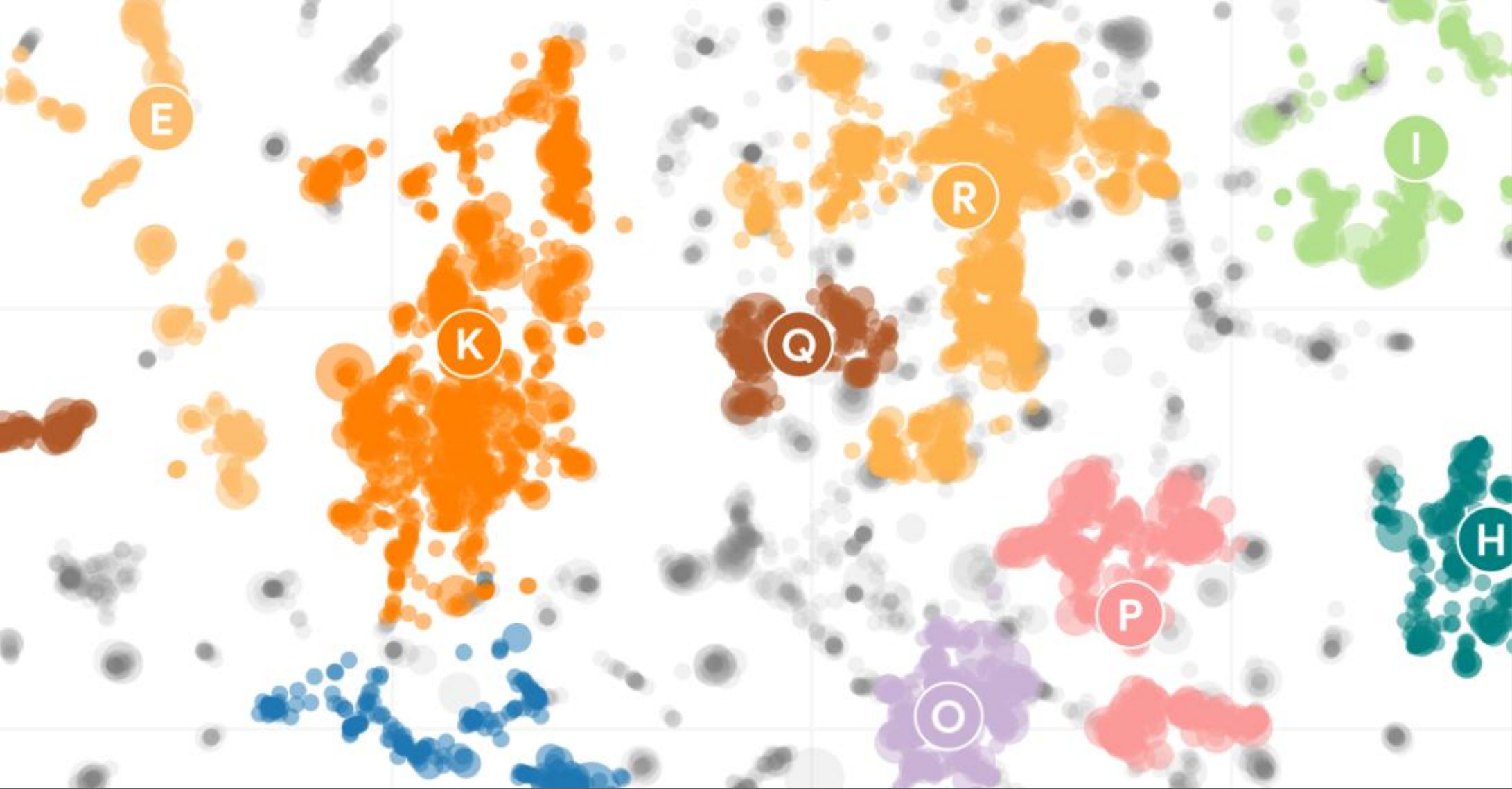
If you compare this map to Savov's, you'd find that there is no "Smith Glitch" here. His approach made so that the authors were being grouped only by their last name in a lot of cases. Because of that, cited "Smiths" were clustered together. His approach made possible to combine data from "Stiny G" and "Stiny George", for example, but had weaknesses. I made some experiments analysing the cited names, and found that if we ignore all but the surnames, the most cited author is "Wang", with 4534 citations in my data. Compare that to the analysis of the names as they come, where "Stiny G" was (a very close) second place with 476 citations, and we can clearly see that its better not to touch it if we don't have a robust approach. "Garbage in, garbage out", I guess...

DATA

The data comes from the Web-of-Science (WOS). I started with the list of publication sources Savov provided, and got whatever was available.

After filtering out the publications, I was left with a list of 15 journals and conferences, that corresponded to about 15 thousand articles and used them to generate the map.





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DISTRIBUTION OF ITEMS BY STANDARD DEVIATION OF THE NUMBER OF TIMES THEY WERE CITED

Mean: 38 | Sigma: 83

Within 0.5 σ (Average): 13418 items

0.5 σ to 2.0 σ (Large): 1582 items

2 σ to 10 σ (Very Large): 360 items

> 10 σ (Extreme): 19 items

THE MAP IS GENERATED FROM DATA FROM THE FOLLOWING PUBLICATIONS:

AAG CONFERENCE PROCEEDINGS (22)

DESIGN STUDIES (698)

ACADIA CONFERENCE PROCEEDINGS (112)

ECADAE CONFERENCE PROCEEDINGS (1909)

AI EDAM (888)

ENVIRONMENT AND PLANNING B-PLANNING & DESIGN (1299)

ARCHITECTURAL DESIGN (579)

INTERNATIONAL JOURNAL OF ARCHITECTURAL COMPUTING (223)

AUTOMATION IN CONSTRUCTION (5949)

NEXUS NETWORK JOURNAL (803)

CAAD FUTURES CONFERENCE PROCEEDINGS (284)

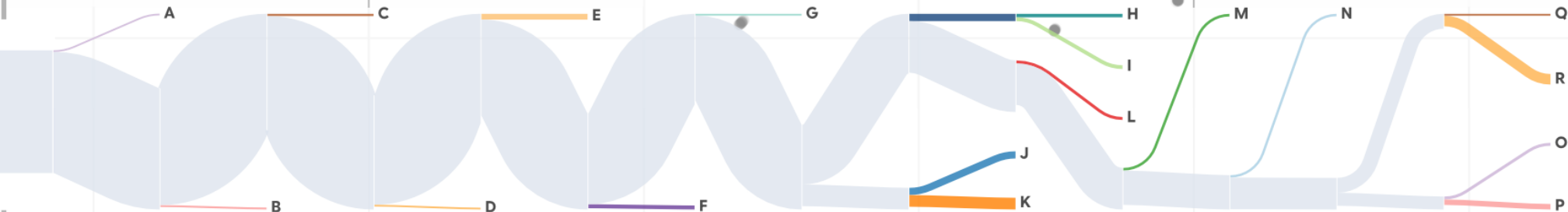
SIMAUD CONFERENCE PROCEEDINGS (105)

CAADRIA CONFERENCE PROCEEDINGS (857)

TECHNE JOURNAL (265)

COMPUTER-AIDED DESIGN (3333)

DATA SNAPSHOT: JUNE 2026



#	CLUSTER TITLE	WORDS & PHRASES	MOST CITED AUTHORS	MOST CITED WORKS
A	CLASSICAL ARCHITECTURE	geometric (74x); perspective (42x); proportions (37x); geometry (35x); mathematical (35x); plan (33x); geometrical (31x); construction (30x); shape (30x); ratios (30x); built (28x); possible (27x); church (26x); palladio (12x)	MARCH,L. (40x); SERLIO,S. (40x); ALBERTI,L. (34x); WITTKOWER,R. (30x); FLETCHER,R. (29x); PALLADIO,A. (28x); VITRUVIUS, (22x); MITROVIC,B. (20x); BARBARO,D. (18x); WILLIAMS,K. (17x)	MARCH L., 1998, ARCHITECTONICS HUMAN (18x); ALBERTI LEONBATTISTA., 1988, ON THE ART OF BUILDING IN TEN BOOKS (14x); PALLADIO ANDREA., 1997, THE FOUR BOOKS ON ARCHITECTURE (14x); HOWARD D., 1982, J SOC ARCHIT HIST, V41, P116, DOI 10.2307/989675 (12x); MITROVIC B., 1990, J SOC ARCHIT HIST, V49, P279, DOI 10.2307/990519 (10x)
B	3D CONCRETE PRINTING	concrete printing dcp (23x); 3d concrete printing (25x); 3d printed concrete (13x); additive manufacturing (55x); concrete printing (34x); construction industry (34x); digital fabrication (25x); future research (23x); 3d printing (39x); 3d printed (16x); construction (258x); printing (239x); concrete (227x); material (167x); materials (97x); printed (94x); fabrication (90x); technology (82x); potential (81x); parameters (77x); structures (75x); manufacturing (71x); structural (69x); 3d (84x); robotic (22x); formwork (14x); automated (12x); scale (12x)	KHOSHNEVIS,B. (86x); BUSWELL,R. (80x); BOS,F. (66x); LIM,S. (61x); MECHTCHERINE,V. (57x); PERROT,A. (55x); WOLFFS,R. (55x); PANDA,B. (47x); GOSSELIN,C. (45x); WANGLER,T. (45x)	KHOSHNEVIS B., 2004, AUTOMAT CONSTR, V13, P5, DOI 10.1016/J.AUTCON.2003.08.012 (53x); LIM S., 2012, AUTOMAT CONSTR, V21, P262, DOI 10.1016/J.AUTCON.2011.06.010 (51x); BUSWELL RA., 2018, CEMENT CONCRETE RES, V112, P37, DOI 10.1016/J.CEMCONRES.2018.05.006 (49x); GOSSELIN C., 2016, MATER DESIGN, V100, P102, DOI 10.1016/J.MATDES.2016.03.097 (45x); ROUSSEL N., 2018, CEMENT CONCRETE RES, V112, P76, DOI 10.1016/J.CEMCONRES.2018.04.005 (37x)
C	AUTOMATED SPATIAL PLANNING	finite element analysis (11x); mesh generation (46x); finite element (20x); mesh adaptation (19x); tetrahedral mesh (18x); mesh quality (17x); mesh (328x); planning (319x); meshes (145x); algorithm (132x); local (90x); quality (79x); surface (72x); urban (70x); elements (63x); boundary (59x); presented (59x); meshing (21x); automatic (14x); environmental (11x); hexahedral (11x)	HEALEY,P. (49x); FALUDI,A. (47x); BLACKER,T. (46x); OWEN,S. (40x); GEORGE,P. (39x); ALEXANDER,E. (32x); SHEWCHUK,J. (32x); FORESTER,J. (32x); INNIS,J. (30x); KNUPP,P. (26x)	BLACKER TD., 1991, INT J NUMER METH ENG, V32, P811, DOI 10.1002/NME.1620320410 (21x); KALBERER F., 2007, COMPUT GRAPH FORUM, V26, P375, DOI 10.1111/J.1467-8659.2007.01060.X (17x); FORESTER J., 1989, PLANNING FACE POWER (14x); OWEN S.J., 1998, P 7 INT MESHING ROUN, P239 (14x); SI H., 2015, ACM T MATH SOFTWARE, V41, DOI 10.1145/2629697 (13x)
D	URBAN SPACE SYNTAX	space syntax (61x); urban design (18x); case study (16x); spatial configurations (16x); spatial configuration (15x); urban (239x); visibility (59x); visual (58x); planning (56x)	HILLIER,B. (169x); BENEDIKT,M. (88x); TURNER,A. (80x); BATTY,M. (61x); PEPONIS,J. (54x); LYNCH,K. (42x); RATTI,C. (30x); PENN,A. (30x); HANSON,J. (26x); GIBSON,J. (26x)	BENEDIKT ML., 1979, ENVIRON PLANN B, V6, P47, DOI 10.1068/B060047 (81x); TURNER A., 2001, ENVIRON PLANN B, V28, P103, DOI 10.1068/B2684 (54x); HILLIER BILL., 1988, THE SOCIAL LOGIC OF SPACE (41x); BATTY M., 2001, ENVIRON PLANN B, V28, P123, DOI 10.1068/B2725 (31x); HILLIER B., 1993, ENVIRON PLANN B, V20, P29, DOI 10.1068/B200029 (29x)
E	AUTOMATED STRUCTURAL	synthetic point clouds (26x); finite element analysis (25x); point cloud data (20x); point cloud segmentation (18x); building information modeling (18x); information modeling bim (13x); element analysis fea (13x); topology optimization method (11x); experimental results demonstrate (11x); real point clouds (11x); point clouds (169x); topology optimization (136x); point cloud (100x); proposed method (74x); deep learning (55x); semantic segmentation	POTTMANN,H. (120x); BENDSOE,M. (115x); WANG,Y. (93x); QI,C. (88x); SIGMUND,O. (70x); J. LIU, Z. ZHANG	QI CR., 2017, PROC CVPR IEEE, P77, DOI 10.1109/CVPR.2017.16 (54x); BENDSOE MP., 1988, COMPUT METHOD APPL M, V71, P197, DOI 10.1016/0045-7825(88)90084-2 (52x); WANG Y., 2019, ACM T GRAPHIC, V38, DOI 10.1145/3326362 (49x);



Thanks for your attention

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This research was supported through the Multiannual Funding of the Landscape, Heritage and Territory Laboratory (Lab2PT), Ref. UID/04509: Laboratório de Paisagens, Património e Território (Lab2PT/UM), financed by national funds (PIDDAC) through the FCT/MCTES, and by an FCT Doctoral Grant with reference 2025.05582.BDANA. The School of Architecture, Art and Design (EAAD) and the Institute of Design of Guimarães (IDEGUI) host and support the Advanced Ceramics R&D Lab and ARENA Lab, offering access to their facilities and equipment necessary for conducting research tests and developing prototypes.